ARTI308 - Machine Learning

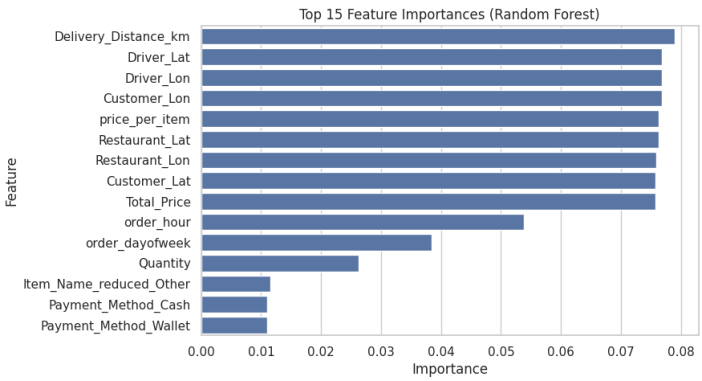
Lab 5: Feature Engineering

This study examines how the value of top_k affects the resulting model's predictive performance.

Three different values of top_k were used for testing: 3, 5, and 6. Each model was trained and evaluated for its performance based on accuracy for each value of top_k. In addition, we also examined the top feature importances to determine which features contributed the most to the resulting model's performance.

Top_k=3:

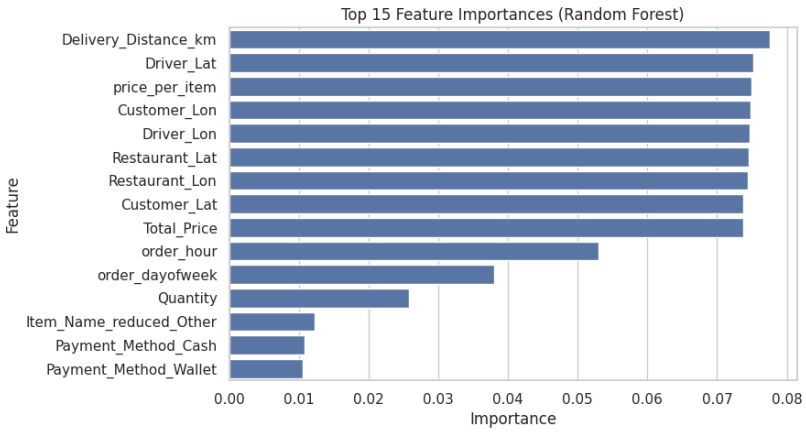
```
[16]
24s
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred), 4))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

Accuracy: 0.4018

Classification Report:
              precision    recall  f1-score   support

   Cancelled       0.40      0.47      0.43      1963
   Delivered       0.41      0.53      0.46      2039
  In Transit       0.21      0.01      0.01       998

    accuracy                           0.40      5000
   macro avg       0.34      0.34      0.30      5000
weighted avg       0.36      0.40      0.36      5000
```



Top_k=5:

```
[35]
26s
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred), 4))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

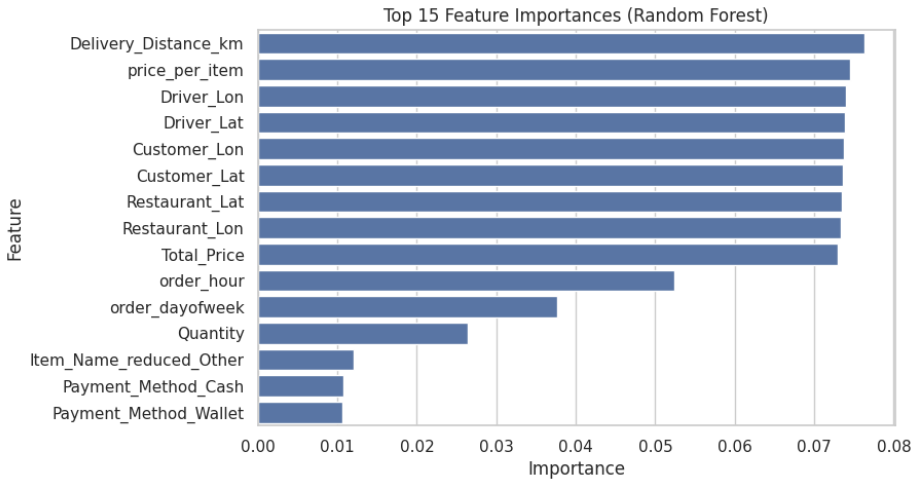
```
Accuracy: 0.3926

Classification Report:
              precision    recall  f1-score   support

   Cancelled       0.39      0.45      0.42      1963
   Delivered       0.40      0.53      0.46      2039
  In Transit       0.08      0.00      0.00       998

    accuracy                           0.39      5000
   macro avg       0.29      0.33      0.29      5000
weighted avg       0.33      0.39      0.35      5000
```



Top_k=6:

```
[55]
23s
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred), 4))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.3908

Classification Report:
              precision    recall  f1-score   support

   Cancelled       0.38      0.46      0.42      1963
   Delivered       0.40      0.52      0.45      2039
  In Transit       0.19      0.00      0.01       998

    accuracy                           0.39      5000
   macro avg       0.32      0.33      0.29      5000
weighted avg       0.35      0.39      0.35      5000
```



Changing the value of top_k affected both the model accuracy and the feature importance.

The highest accuracy 0.4018, was achieved when top_k = 3. As the value of top_k increased to 5 and 6, the accuracy slightly decreased.

This suggests that using fewer item categories helps simplify the model and improves its performance. In contrast, increasing the number of categories may introduce additional noise, which can negatively affect accuracy.

Across all experiments, the most important features remained consistent.

On the other hand, item-related features had minimal impact on the model's predictions, regardless of the value of top_k.